LPS
APP
NADPH oxidase complex
JNK
NOS2
FOS
JUNB
AP-1 transcription factor
nucleus
SRR
nitric oxide
SNP
SNAP
dexamethasone
SRR steady-state dimer
inactive SDS/βME-resistant SRR dimer
peroxynitrite
Spermine NONOate
HAPI microglia
NMDA glutamate receptor activity
neuron
IL1B
APP
IL4
APOE
PKA
dibutyryl cAMP
activated microglia M1
ARG1
CHI3L1
SLC7A11
microglia
L-glutamate
AP5
NMDA receptor glycine/D-serine site
DCKA
NMDA glutamate receptor activity
L-NIO
NOS3
20-HETE
vasodilation
D-serine
NP-EGTA
ACPD
Metabotropic Glutamate Receptor
Ca2+
prostaglandin E2
astrocyte activation
GLUD1
GLUD2
glutamate dehydrogenase
glutamate catabolic process
D-glyceraldehyde 3-phosphate
NAD(+)
NADH
MPP+
GAPDH
ATP
glycolysis
ER stress response
neuron death
nucleocytoplasmic shuttling
D-serine biosynthetic process
DAO
BME
etoposide
GAPDH
LMNA
SRR : LMNA : GAPDH
isoform 1
FBXO22
SRR : FBXO22a
astrocyte
6-OHDA
negative regulation of oxidative stress-induced neuron death
APP
IL4
neuron death
L-glutamate
SYP
protofibril
dibutyryl cAMP
cAMP
cyclic nucleotide phosphodiesterase
IBMX
caffeine
PHGDH
PHGDH tetramer
PSAT1
pyridoxal 5'-phosphate
PSAT1 : PXLP dimer
NAD(+)
NADH
PSPH
Mg2+
PSPH : Mg2+ dimer
L-serine biosynthetic process
microcephaly
SLC1A4
SRR : pyridoxal 5'-phosphate
neuron development
axon guidance
striatal neuron
striatum
serine racemase activity
pyruvate dehydrogenase complex
mitochondrial matrix
BAX
apoptosis
staurosporine
DA neuron
substantia nigra
astrocyte activation
NMDA receptor
KCl
late phase of apoptosis
dizocilpine
met-phen
MAPK8
MAPK9
MAPK10
JNKs
UPS
granule neuron
cerebellum
corpus callosum
pyramidal neuron
SRR : PICK1
PICK1
PRKCA
EPHB3
PICK1 : EPHB3
EPHA4
PICK1 : EPHA4
EFNB3
astrocyte
SNARE
vesicle fusion
astrocytic vesicle
D-serine
SLC7A10
glycine
antiporter activity
long-term synaptic potentiation
NMDA glutamate receptor activity
NMDA receptor
long-term synaptic potentiation
induction of synaptic plasticity by chemical substance
neuron
CA1
brain

SRR
GLUD1
GLUD2
glutamate dehydrogenase
V-ATPase
ER stress response
DAO
nucleocytoplasmic shuttling
FBXO22
AP5
ACPD
BME
TGFB1
NP-EGTA
glutamate catabolic process
H+
Metabotropic Glutamate Receptor
neuron death
ER stress response
L-glutamate
voltage-gated sodium channel acti
caffeine
IBMX
etoposide
glycine
MK-801
EFNB3
LPS
APP
Spermine NONOate
NADPH oxidase complex
SNP
NOS2
SNAP
6-OHDA
MPP+
proton transmembrane transport
Ca2+
PHGDH
MPP+
GAPDH
nucleocytoplasmic shuttling
neuron death
cyclic nucleotide phosphodiest
NMDA receptor
dibutyryl cAMP
IL4
PHGDH
PHGDH
PHGDH
PHGDH
PHGDH tetramer
PSPH
Mg2+
PSPH : Mg2+ dimer
PSAT1
PSAT1
pyridoxal 5'-phosphate
pyridoxal 5'-phosphate
PSAT1 : PXLP dimer
PICK1
EPHA4
PICK1 : EPHA4
PICK1
EPHB3
PICK1 : EPHB3
PRKCA
S657
P
ATP
PRKCA
EPHB3
EPHA4
SNARE
met-phen
UPS
JNK
dexamethasone
nitric oxide
peroxynitrite
neuron death
Mn2+
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
positive regulation of oxidative stress-induced neuron death
synaptic vesicle lumen acidification
SRR
Cl-
SLC1A4
astrocyte activation
L-serine biosynthetic process
glycolysis
D-serine biosynthetic process
glyceraldehyde 3-phosphate
TGFB receptor signaling pathway
NADH
NAD(+)
SRR
FBXO22
SRR : FBXO22
neuron
PRKCA
cAMP
synaptogenesis
dibutyryl cAMP
PKA
IL1B
APP
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
SRR
SRR
Mg2+
PRKCA
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AP-1 transcription factor
SRR
SRR
SRR
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SRR
SRR
SRR steady-state dimer
astrocyte activation
serine racemase activity
staurosporine
pyruvate dehydrogenase complex
BCL2
T69
S70
S80
S89
BAX
SRR
D-serine
axon guidance
microcephaly
neuron development
13J
prostaglandin E2
D-serine
mortor activity
GAPDH
LMNA
SRR
SRR : LMNA : GAPDH
isoform 1
FBXO22
SRR
SRR
PKA
ARG1
SLC7A11
CHI3L1
APOE
D-serine
SRR
PICK1
SRR : PICK1
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NMDA receptor glycine/D-serine site
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SLC7A10
apoptosis
NMDA glutamate receptor activ
L-NIO
MAPK8
T183
Y185
MAPK9
MAPK10
JNKs
late phase of apoptosis
NOS3
20-HETE
vasodilation
antiporter activity